1. Parts order form

Assumptions: Assume one order per customer at a recorded date/time, and one row per part within that order.
customerNo fixes the customer name/type; partNo fixes the name, type, single cage and catalog price (no order-specific prices).
Employees may serve any customer; partType does not determine cageCode. The business heading is fixed form text.
All 12 original fields are retained; labels are clarified and US dates are shown as ISO dates. No identifier or other field is added.

Unnormalized form

order_form

customerNo	customerName	customerType	**orderDate**	**orderTime**	employee	partNo	partName	partType	cageCode	quantityOrdered	unitPrice
HG54587	Jeff Peterson	Consumer	2024-07-01	10:30	D. Harrison	10654 10456 10776 10657	Float Control Modulator Hose Assembly Float Assembly	Plumbing Electrical Plumbing Plumbing	G413 H433 G413 G413	4 3 7 5	12 7 9 10

Record identifier: (customerNo, orderDate, orderTime). The six aligned part lists are a repeating group, so this is UNF.

1NF: Expand the parts into four rows

Repeat the order fields for each part; add the existing partNo to the composite PK so each line is uniquely identified.

order_entry

customerNo	customerName	customerType	**orderDate**	**orderTime**	employee	**partNo**	partName	partType	cageCode	quantityOrdered	unitPrice
HG54587	Jeff Peterson	Consumer	2024-07-01	10:30	D. Harrison	10654	Float Control	Plumbing	G413	4	12
HG54587	Jeff Peterson	Consumer	2024-07-01	10:30	D. Harrison	10456	Modulator	Electrical	H433	3	7
HG54587	Jeff Peterson	Consumer	2024-07-01	10:30	D. Harrison	10776	Hose Assembly	Plumbing	G413	7	9
HG54587	Jeff Peterson	Consumer	2024-07-01	10:30	D. Harrison	10657	Float Assembly	Plumbing	G413	5	10

Not 2NF: customer facts depend only on customerNo, part facts only on partNo, and employee only on the order key.

2NF: Remove partial dependencies to get 2NF

customerNo -> customerName, customerType; partNo -> partName, partType, cageCode, unitPrice.
(customerNo, orderDate, orderTime) -> employee; the full line key -> quantityOrdered.

customer

customerNo
customerName
customerType

part

partNo
partName
partType
cageCode
unitPrice

order_header

customerNo
orderDate
orderTime
employee

order_line

customerNo
orderDate
orderTime
partNo
quantityOrdered

3NF: Sort into tables

These 2NF relations already have no transitive non-key dependency under the assumptions; keep all four and mark their FKs below.

customer

customerNo	customerName	customerType
HG54587	Jeff Peterson	Consumer

part

partNo	partName	partType	cageCode	unitPrice
10654	Float Control	Plumbing	G413	12
10456	Modulator	Electrical	H433	7
10776	Hose Assembly	Plumbing	G413	9
10657	Float Assembly	Plumbing	G413	10

order_header

customerNo	**orderDate**	**orderTime**	employee
HG54587	2024-07-01	10:30	D. Harrison

order_line

customerNo	***orderDate***	***orderTime***	***partNo***	quantityOrdered
HG54587	2024-07-01	10:30	10654	4
HG54587	2024-07-01	10:30	10456	3
HG54587	2024-07-01	10:30	10776	7
HG54587	2024-07-01	10:30	10657	5

2. Therapist appointments

Assumptions: staffNo and patNo uniquely identify a therapist and patient; names need not be unique.
A therapist sees one patient at a given date/time and works at one branch per day; the branch may change on other days.
Patients may have multiple appointments per day with different therapists.

1NF: Starting table

All values are atomic, with one row per appointment and no repeating group.

therapist_appointment

staffNo	therapistName	patNo	patName	**appointmentDate**	**appointmentTime**	branchNo
S1011	Fred Smith	P100	Lily White	2022-09-12	10:00	M15
S1011	Fred Smith	P105	Jill Baker	2022-09-12	12:00	M15
S1024	Heidi Pierce	P108	Andy McKee	2022-09-12	10:00	Q10
S1024	Heidi Pierce	P108	Andy McKee	2022-09-14	14:00	Q10
S1032	Richard Levin	P105	Jill Baker	2022-09-14	16:30	M15
S1032	Richard Levin	P110	Jimmy Winter	2022-09-15	18:00	B13

Why it is not 2NF

- staffNo -> therapistName depends on only part of the appointment key.
- (staffNo, appointmentDate) -> branchNo also depends on only part of the key.

2NF: Remove partial dependencies

Move therapistName to therapist and branchNo to therapist_schedule.
Keep patNo and patName in appointment; patNo -> patName is still a transitive dependency.

therapist

staffNo
therapistName

therapist_schedule

staffNo
appointmentDate
branchNo

appointment

staffNo
appointmentDate
appointmentTime
patNo
patName

3NF: Sort into tables

Move patName to patient using patNo -> patName to remove the transitive dependency; retain patNo in appointment as an FK.

therapist

staffNo	therapistName
S1011	Fred Smith
S1024	Heidi Pierce
S1032	Richard Levin

patient

patNo	patName
P100	Lily White
P105	Jill Baker
P108	Andy McKee
P110	Jimmy Winter

therapist_schedule

staffNo	**appointmentDate**	branchNo
S1011	2022-09-12	M15
S1024	2022-09-12	Q10
S1024	2022-09-14	Q10
S1032	2022-09-14	M15
S1032	2022-09-15	B13

appointment

staffNo	***appointmentDate***	**appointmentTime**	*patNo*
S1011	2022-09-12	10:00	P100
S1011	2022-09-12	12:00	P105
S1024	2022-09-12	10:00	P108
S1024	2022-09-14	14:00	P108
S1032	2022-09-14	16:30	P105
S1032	2022-09-15	18:00	P110

3. Event staff

Assumptions: eNo, contractNo and eventNo uniquely identify an employee, contract and event.
Each contract covers one event; an event has one location and may have several contracts.
Each employee-contract pair occurs once; hours records the total hours for that pair.

1NF: Starting table

All values are atomic; the composite key (eNo, contractNo) identifies each employee-contract pair.

event_staff

eNo	**contractNo**	hours	eName	eventNo	eventLoc
1135	C1024	16	Smith J	H25	Queens
1057	C1024	24	Hocine D	H25	Queens
1068	C1025	28	White T	H4	Yonkers
1135	C1025	15	Smith J	H4	Yonkers
1135	C1026	10	Smith J	H25	Queens

Why it is not 2NF

- eNo: eName depends on only one part of the key.
- contractNo: eventNo, eventLoc also depends on only one part of the key.

2NF: Remove partial dependencies

Move eName to employee and eventNo/eventLoc to contract.
Keep hours in contract_assignment because it depends on both eNo and contractNo.

employee

eNo
eName

contract

contractNo
eventNo
eventLoc

contract_assignment

eNo
contractNo
hours

3NF: Sort into tables

Move eventLoc to event using eventNo -> eventLoc to remove the transitive dependency; retain eventNo in contract as an FK.

employee

eNo	eName
1135	Smith J
1057	Hocine D
1068	White T

event

eventNo	eventLoc
H25	Queens
H4	Yonkers

contract

contractNo	*eventNo*
C1024	H25
C1025	H4
C1026	H25

contract_assignment

eNo	***contractNo***	hours
1135	C1024	16
1057	C1024	24
1068	C1025	28
1135	C1025	15
1135	C1026	10